Amar Ali-bey

Montréal, Canada — [*Open to relocation*]
amar.alibey@gmail.com | amaralibey | amaralibey | (+1) 418-900-8418
https://amaralibey.github.io

ABOUT

- **Research and Publications** — Published in top AI venues: CVPR, BMVC, WACV, Neurocomputing.
- **Open-Source Contributions** — Created and maintained open-source projects such as OpenVPRLab and GSV-Cities (+1600 downloads, 500+ GitHub stars).
- **Deep Learning Expertise** — Trained 100+ models for Image Retrieval, Face Recognition, Semantic Segmentation, Object Detection.
- **Project Management** — Autonomous in full project lifecycle: data collection to model deployment.
- **Communication** — Effectively present complex technical concepts at conferences and team meetings.

EDUCATION

Ph.D. in Computer Vision and Machine Learning 2017 — 2024
Laval University, Québec, Canada
Thesis title: Deep Representation Learning for Visual Place Recognition.

- Advisors: Prof. Brahim Chaib-draa and Prof. Philippe Giguère.

Master's in Computer Science 2015 — 2017
Laval University, Québec, Canada
Took advanced courses on Mobile Robotics, Machine Learning and Optimization. Explored the following research fields: Image Retrieval , Object Detection and Semantic Segmentation.
Transitioned directly into Ph.D.

Student Internship 2013 — 2014
University of Technology of Compiègne (UTC), Compiègne, France
Worked on the Vehicle Routing Problem with Time Windows (VRPTW), which consists of selecting the optimal routes for a fleet of vehicles to service customers within specific time frames.

- Advisors: Prof. Aziz Moukrim and Prof. Sohaib Afifi.

B.S. in Software Engineering (with honors) 2009 — 2014
Ecole nationale Supérieure d'Informatique (ESI), Algiers, Algeria
Attended Algeria's most competitive engineering school, and graduated as **Valedictorian**.

FIRST AUTHOR PUBLICATIONS

BoQ: A Place is Worth a Bag of Learnable Queries (CVPR 2024)
Amar Ali-bey, Brahim Chaib-draa and Philippe Giguère

- Bag-of-Queries is a new architecture for visual place recognition.
- Achieves state-of-the-art on all existing benchmarks, with big margins.
- Paper, code, and trained models: `https://github.com/amaralibey/Bag-of-Queries`

MixVPR: Feature Mixing for Visual Place Recognition (WACV 2023)
Amar Ali-bey, Brahim Chaib-draa and Philippe Giguère

- MixVPR is a highly influential technique, it has been cited more than 60 times in its first year.
- Paper, code, and trained models: `https://github.com/amaralibey/MixVPR`

Global Proxy-based Hard Mining for Visual Place Recognition (BMVC 2022)
Amar Ali-bey, Brahim Chaib-draa and Philippe Giguère

- Introduced global informative mini-batch sampling based on proxies.
- Paper and video presentation: `https://bmvc2022.mpi-inf.mpg.de/958/`

GSV-Cities: Toward Appropriate Supervised Visual Place Recognition (Neurocomputing 2021)
Amar Ali-bey, Brahim Chaib-draa and Philippe Giguère

- Introduced **GSV-Cities**, a new large-scale dataset for Visual Place Recognition.
- Implemented a novel training framework leveraging this dataset.
- Introduced **Conv-AP**, a new fully convolutional architecture achieving state new state-of-the-art.
- Framework: `https://github.com/amaralibey/gsv-cities`
- Dataset: `https://kaggle.com/datasets/amaralibey/gsv-cities`

ACADEMIC ACHIEVEMENTS AND ACTIVITIES

1st Rank Awards 2012, 2013, 2014
Received annual prizes from DJEZZY, a school partner, for maintaining top of class.

4th Place in Nationwide Competition 2014
National scholarship competition for doctoral studies, open only to top graduates of Algerian universities.

Attended International Computer Vision Summer School (ICVSS) 2017

Volunteered at IEEE International Conference on Robotics and Automation (ICRA) 2019

Presented at Montreal AI Symposium 2022

Reviewed papers for many AI venues (ICRA, IROS, WACV, RA-L) 2020-2024

WORK EXPERIENCE

Graduate Teaching Assistant 2017 – 2023
Laval University, Quebec, Canada

Prepared new materials for courses, assisted students with technical questions, and taught parts of the courses:

- **Advanced Techniques In Artificial Intelligence (IFT-4102/7025)** [100+ students]
- **Deep Learning (GLO-4030/7030)** [70+ students]
- **Practical Machine Learning (GLO-7050)** [50+ students]
- **Introduction to Mobile Robotics (GLO-4001/7021)** [80+ students]
- **Exam and Homework correction**
  - Introduction to Mobile Robotics (GLO-4001/7021)
  - Advanced Programming In C ++ (GIF-1003)
  - Introduction to Programming (IFT-1004)
  - Mathematics for Computer Science (MAT-1919)

Deep Learning Consultant 2018 – 2024
Short/Medium term interventions with high impact on performance.

- Football players Re-Identification in video streams
- Traffic cone detection
- Semantic segmentation for urban scenes
- Helmet detection for safety in work sites
- Enhanced the deep learning pipeline for object tracking and counting

The Autonomous Vehicle of Laval University (VAUL) 2018

- Developed a cone detector using OpenCV to enhance the annotation of cones in circuit videos.
- Trained YOLO models for real-time cone detection in video streams.

PROJECTS

OpenVPRLab https://github.com/amaralibey/OpenVPRLab

This is a comprehensive open-source framework for visual place recognition (VPR). It accelerates the training and development of deep models for VPR, providing an easy-to-use platform for researchers and developers to achieve *reproducible* state-of-the-art performance. Key features include:

- Highly modular design allowing focused development on specific VPR components.
- Easy dataset management with a single-command download for training and validation datasets.
- Extensive loss functions integration from the deep metric learning community.
- Integrated Tensorboard support for tracking experiments.
- Performance visualization tools for analyzing model retrievals.
- Flexibility to develop custom aggregators, backbones, or loss functions.

GSV-Cities https://www.kaggle.com/datasets/amaralibey/gsv-cities

A large-scale dataset for training deep neural networks for Visual Place Recognition (VPR). The dataset contains more than 500k images depicting over 60k different places spread across multiple global cities. Each place is represented by 4 up to 20 images, ensuring comprehensive coverage and diversity. All places are physically distant, providing full supervision for VPR tasks.

SKILLS

Expert level:

- Python, PyTorch, Numpy, LightningAI, FAISS, TensorBoard, OpenCV
- Image-based Localization, Image Retrieval, Similarity Search, Contrastive Learning
- Designing and maintaining deep learning pipelines

Good level:

- C, C++, PHP, SQL, HTML/CSS
- Face Recognition, Person Re-identification, Semantic Segmentation, Object Detection, Image Classification, NLP

Languages

- **French** : Native speaker
- **English** : Fluent

REFERENCES

Prof. Brahim Chaib-draa (brahim.chaib-draa@ift.ulaval.ca)
Prof. Philippe Giguère (philippe.giguere@ift.ulaval.ca)